MIL-C-62/7B
19 December 1968
SUPERSEDING
62/7A
25 November 1964

# MILITARY SPECIFICATION SHEET

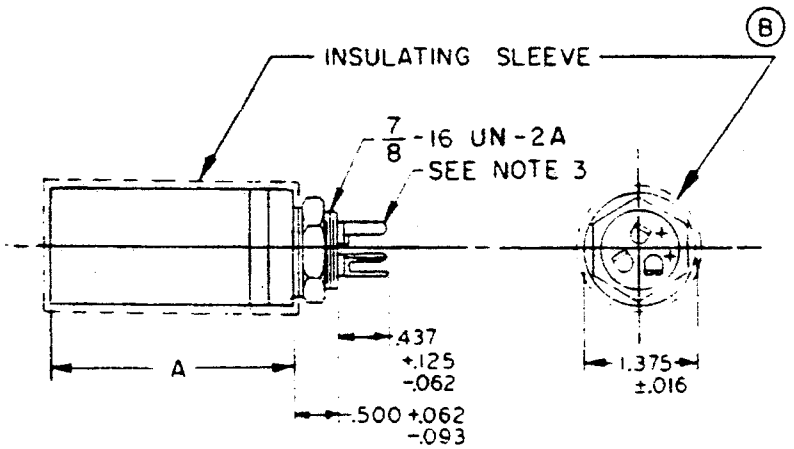
## CAPACITORS, FIXED, ELECTROLYTIC

## (DC, ALUMINUM, DRY ELECTROLYTE, POLARIZED)

## STYLES CE42 AND CE45

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-62.

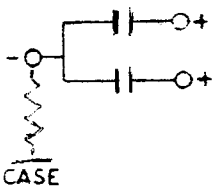
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.




| INCHES | MM |
|---|---|
| .016 | .41 |
| .062 | 1.57 |
| .093 | 2.36 |
| .125 | 3.18 |
| .437 | 11.10 |
| .500 | 12.70 |
| 1.375 | 34.93 |

STYLES
CE42 – uninsulated
CE45 – insulated

UNINSULATED STYLE NOT FOR NEW DESIGN.
FOR REPLACEMENT PURPOSES ONLY.




NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Shape and position of lugs optional.
4. Mounting nut and lockwasher shall be supplied with each capacitor (Family 3).
5. There is an indeterminate resistance between the metal case and the negative terminal.

Ⓑ denotes changes

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FSC 5910

(B) TABLE I. Styles CE42 and CE45 capacitors.

| Type designation [1] | Capacitance | DC rated voltage per section | DC leakage per section | Maximum impedance per section | Dissipation factor per section | Case dimensions [2] A [3] ± .062 (1.57) |
|---|---|---|---|---|---|---|
| | uf | volts | ua | ohms | percent | inches |
| CE4-C330J | 33-33 | 150 | 106 | 266 | 12 | 2.250 (57.15) |
| CE4-C680J | 68-68 | 150 | 151 | 129 | 12 | 2.250 (57.15) |
| CE4-C750J | 75-75 | 150 | 160 | 117 | 12 | 2.250 (57.15) |
| CE4-C151J | 150-150 | 150 | 225 | 59 | 12 | 3.000 (76.20) |
| CE4-C221J | 220-220 | 150 | 273 | 40 | 12 | 3.750 (95.15) |
| CE4-C620K | 62-62 | 200 | 166 | 142 | 12 | 2.250 (57.15) |
| CE4-C680K | 68-68 | 200 | 175 | 129 | 12 | 2.500 (63.50) |
| CE4-C750K | 75-75 | 200 | 183 | 117 | 12 | 2.500 (63.50) |
| CE4-C101K | 100-100 | 200 | 211 | 88 | 12 | 2.750 (69.85) |
| CE4-C151K | 150-150 | 200 | 260 | 59 | 12 | 3.250 (82.55) |
| CE4-C300M | 30-30 | 250 | 129 | 293 | 12 | 2.250 (57.15) |
| CE4-C470M | 47-47 | 250 | 166 | 187 | 12 | 2.250 (57.15) |
| CE4-C560M | 56-56 | 250 | 178 | 157 | 12 | 2.500 (63.50) |
| CE4-C680M | 68-68 | 250 | 195 | 129 | 12 | 2.750 (69.85) |
| CE4-C750M | 75-75 | 250 | 206 | 117 | 12 | 3.000 (76.20) |
| CE4-C101M | 100-100 | 250 | 237 | 88 | 12 | 3.250 (82.55) |
| CE4-C360N | 36-36 | 300 | 156 | 244 | 12 | 2.250 (57.15) |
| CE4-C680N | 68-68 | 300 | 214 | 129 | 12 | 3.250 (82.55) |
| CE4-C101N | 100-100 | 300 | 260 | 88 | 12 | 4.250 (107.95) |
| CE4-C100P | 10-10 | 350 | 88 | 880 | 12 | 2.250 (57.15) |
| CE4-C200P | 20-20 | 350 | 125 | 440 | 12 | 2.250 (57.15) |
| CE4-C300P | 30-30 | 350 | 154 | 293 | 12 | 2.250 (57.15) |
| CE4-C470P | 47-47 | 350 | 192 | 187 | 12 | 2.750 (69.85) |
| CE4-C620P | 62-62 | 350 | 222 | 142 | 12 | 3.250 (82.55) |
| CE4-C680P | 68-68 | 350 | 239 | 129 | 12 | 3.750 (95.25) |

See footnotes at end of table.



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(B) TABLE I. Styles CE42 and CE45 capacitors. (Continued)

| Type designation 1/ | Capacitance | DC rated voltage per section | DC leakage per section | Maximum impedance per section | Dissipation factor per section | Case dimensions 2/ A 3/ ± .062 (1.57) |
|---|---|---|---|---|---|---|
| | uf | volts | ua | ohms | percent | inches |
| CE4-C100Q | 10-10 | 400 | 94 | 1480 | 17 | 2.250 (57.15) |
| CE4-C150Q | 15-15 | 400 | 116 | 986 | 17 | 2.250 (57.15) |
| CE4-C220Q | 22-22 | 400 | 141 | 670 | 17 | 2.250 (57.15) |
| CE4-C330Q | 33-33 | 400 | 172 | 450 | 17 | 3.000 (76.20) |
| CE4-C470Q | 47-47 | 400 | 206 | 314 | 17 | 3.500 (88.90) |
| CE4-C100R | 10-10 | 450 | 100 | 2200 | 17 | 2.250 (57.15) |
| CE4-C150R | 15-15 | 450 | 123 | 1465 | 17 | 2.250 (57.15) |
| CE4-C220R | 22-22 | 450 | 149 | 1000 | 17 | 2.750 (69.85) |
| CE4-C300R | 30-30 | 450 | 174 | 733 | 17 | 3.500 (88.90) |
| CE4-C430R | 43-43 | 450 | 208 | 512 | 17 | 3.750 (95.25) |

1/ Complete type designation shall include an additional digit to complete style.

2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

3/ The dimensions shown are for style CE42; for style CE45, add .062 (1.57) ± .062 (1.57) to the nominal dimension.



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REQUIREMENTS:

Dimensions and configuration:
See figure and table I.

Case type: Tubular, uninsulated (CE42) and insulated (CE45).

Capacitance value: See table I.

Capacitance tolerance:
At 200 volts, dc rated, and above:
-10, +50 percent.
At 150 volts, dc rated, and less:
-10, +75 percent.

Operating temperature range:
-40° C to +85° C.

(B) Solderability: Method 208 of MIL-STD-202.

DC leakage: See table I.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor: See table I.

Terminal strength: Method 211 of MIL-STD-202, test conditions A, B, and E.

Shock, medium impact: Method 205, test condition C, MIL-STD-202.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Temperature cycling and immersion:
Temperature cycling: Method 102, test condition D, MIL-STD-202.
Immersion: Method 104, test condition C, MIL-STD-202.
DWV-Insulating sleeves-Method 301 of MIL-STD-202. 4,000 volts dc applied.
DCL - Not more than 150 percent of initial measurement.
Capacitance change - 5 percent.
DF - See table I.
Corrosion - Not more than 10 percent.

Moisture resistance: Method 106 of MIL-STD-202.
Insulation resistance (insulating sleeve) - Not less than 100 megohms.
DWV-Insulating sleeves-Method 301 of MIL-STD-202. 4,000 volts dc applied.
DCL - Not more than 150 percent of initial requirement.
Capacitance change - 5 percent.
DF - See table I.
Corrosion - Not more than 10 percent.

Stability at low and high temperature:
In accordance with MIL-C-62.
Step 1 (at +25° C):
DCL - See table I.
Capacitance - See table I.
DF - See table I.

Step 2 (at -40° C):
Impedance - See table I.

Step 3 (at +25° C):
DCL - See table I.
Capacitance - ±5 percent of initial value.
DF - See table I.

Step 4 (at +85° C):
DCL - Not more than 10 times initial value, or meet initial 25° C requirement.
Capacitance - ±20 percent of initial value.
DF - See table I.

Step 5 (at +25° C):
DCL - See table I.
Capacitance - ±5 percent of initial value.
DF - See table I.

Surge voltage: In accordance with MIL-C-62.
DCL - See table I.
Capacitance change - 6 percent.
DF - See table I.

Vent: Not applicable.

Life: Method 108, test condition F, MIL-STD-202.
DCL - See table I.
Capacitance change:
100 volts, dc rated, and less:
-20, +15 percent.
DC rated voltage above 100 volts:
±15 percent.
DF - Meet the initial requirement or not exceed 150 percent of the initial measurement.

Barometric pressure: Method 105 of MIL-STD-202.

Shelf life: In accordance with MIL-C-62.
Build factor: Not more than 4.0.
Shelf factor: Not more than 3.0.
DCL: Not more than 150 percent of initial value.
Capacitance change: 6 percent.
DF: Not more than 175 percent of initial requirement.

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Ⓑ TABLE II. Supersession data.

| Superseded insulated part number | Superseded uninsulated part number | Recommended replacement style CE42 or CE45 | Superseded insulated part number | Superseded uninsulated part number | Recommended replacement style CE42 or CE45 |
|---|---|---|---|---|---|
| 001 | 078 | CE4-C100P | 027 | 104 | CE4-C101M |
| 002 | 079 | CE4-C300M | 028 | 105 | CE4-C100P |
| 003 | 080 | CE4-C330J | 029 | 106 | CE4-C300P |
| 004 | 081 | CE4-C470M | 030 | 107 | CE4-C360N |
| 005 | 082 | CE4-C680J | 031 | 108 | CE4-C470P |
| 006 | 083 | CE4-C750J | 032 | 109 | CE4-C620P |
| 007 | 084 | CE4-C101K | 033 | 110 | CE4-C680N |
| 008 | 085 | CE4-C151J | 034 | 111 | CE4-C101N |
| 009 | 086 | CE4-C221J | 035 | 112 | CE4-C100P |
| 010 | 087 | CE4-C100P | 036 | 113 | CE4-C200P |
| 011 | 088 | CE4-C300M | 037 | 114 | CE4-C300P |
| 012 | 089 | CE4-C360N | 038 | 115 | CE4-C470P |
| 013 | 090 | CE4-C470M | 039 | 116 | CE4-C470P |
| 014 | 091 | CE4-C470M | 040 | 117 | CE4-C620P |
| 015 | 092 | CE4-C620K | 041 | 118 | CE4-C680P |
| 016 | 093 | CE4-C680K | 042 | 119 | CE4-C100Q |
| 017 | 094 | CE4-C750K | 043 | 120 | CE4-C150Q |
| 018 | 095 | CE4-C101K | 044 | 121 | CE4-C220Q |
| 019 | 096 | CE4-C151K | 045 | 122 | CE4-C330Q |
| 020 | 097 | CE4-C100P | 046 | 123 | CE4-C470Q |
| 021 | 098 | CE4-C300M | 047 | 124 | CE4-C100R |
| 022 | 099 | CE4-C300M | 048 | 125 | CE4-C150R |
| 023 | 100 | CE4-C470M | 049 | 126 | CE4-C220R |
| 024 | 101 | CE4-C560M | 050 | 127 | CE4-C300R |
| 025 | 102 | CE4-C680M | 051 | 128 | CE4-C430R |
| 026 | 103 | CE4-C750M | | | |

Application data:

All capacitors shall withstand an rms ripple current at 120 Hz and 85° C as determined from the formula:

$$I_r = k\sqrt{C}$$

Where:

$I_r$ = rms ripple current in milliamps.
C = nominal capacitance in microfarads per section.
k = 50 for units rated at 150 and 200 volts dc.
= 35 for units rated at 250 to 350 volts dc. incl.
= 30 for units rated at 400 and 450 volts dc.

Ⓑ Supersession data:

See table II.



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Custodians:
Army - EL
Navy - EC
Air Force 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

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